

Analytic function

Analytic function:

A function $f(z)$ is said to be analytic at the point $z=z_0$ if it is single valued and differentiable at that point.

Derivative of $f(z) = u + iv$:

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

Cauchy Riemann equation:

The Cauchy Riemann equation for the function $f(z) = u + iv$ is

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

problems:

Show that the function $f(z) = e^x (\cos y + i \sin y)$ is analytic, also find its derivative.

Solution:

The function $f(z)$ will be analytic if $f(z)$ satisfies the Cauchy Riemann condition $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

Given that

$$f(z) = e^x (\cos y + i \sin y)$$

$$\Rightarrow f(z) = e^x \cos y + i e^x \sin y$$

$$\text{Let } u + iv = f(z) = e^x (\cos y + i \sin y)$$

$$\Rightarrow u + iv = f(z) = e^x \cos y + i e^x \sin y$$

Separating the real and imaginary part

$$u = e^x \cos y \quad \text{and} \quad v = e^x \sin y$$

$$\therefore \frac{\partial u}{\partial x} = e^x \cos y$$

$$\therefore \frac{\partial v}{\partial x} = e^x \sin y$$

$$\Rightarrow -\frac{\partial v}{\partial x} = -e^x \sin y$$

$$\therefore \frac{\partial u}{\partial y} = e^x (-\sin y)$$

$$\therefore \frac{\partial v}{\partial y} = e^x \cos y$$

$$\Rightarrow \frac{\partial u}{\partial y} = -e^x \sin y$$

$$\text{Since } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

So the given function $f(z) = e^x (\cos y + i \sin y)$ is analytic.

Now, ~~f(z)~~ the derivative of $f(z)$ is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= e^x \cos y + i e^x \sin y$$

$$\Rightarrow f'(z) = e^x (\cos y + i \sin y)$$

(Ans:)

problem:

Check the function $f(z) = z \operatorname{Re}\{z\}$ is analytic or not, also find its derivative.

Solution:

The function $f(z)$ will be analytic if $f(z)$ satisfies Cauchy Riemann condition

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\text{Let } u + iv = f(z) = z \operatorname{Re}\{z\}$$

$$= (x + iy)x \quad \left[\begin{array}{l} \because z = x + iy \\ \operatorname{Re}\{z\} = x \end{array} \right]$$

$$\Rightarrow u + iv = f(z) = x^2 + ixy$$

Separating real and imaginary part

$$u = x^2 \quad \text{and} \quad v = xy$$

$$\therefore \frac{\partial u}{\partial x} = 2x$$

$$\therefore \frac{\partial v}{\partial x} = 1 \cdot y = y$$

$$\Rightarrow -\frac{\partial v}{\partial x} = -y$$

$$\therefore \frac{\partial u}{\partial y} = 0$$

$$\therefore \frac{\partial v}{\partial y} = x \cdot 1 = x$$

$$\text{Since } \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x}$$

so the given function $f(z) = z \operatorname{Re}\{z\}$ is ^{not} analytic.

Now, the derivative of $f(z)$ is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ = 2x + iy$$

$$\Rightarrow f'(z) = 2x + iy \text{ (Ans.)}$$

problem:

Check the function $f(z) = z^2$ is analytic or not, also find its derivative.

Solution:

The function $f(z)$ will be analytic if $f(z)$ satisfies Cauchy Riemann condition

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\text{Let } u + iv = f(z) = z^2$$

$$= (x + iy)^2 \quad [\because z = x + iy]$$

$$= x^2 + 2x \cdot iy + i^2 y^2$$

$$= x^2 + 2xyi + (-1)y^2 \quad [\because i^2 = -1]$$

$$= x^2 + 2xyi - y^2$$

$$\Rightarrow u + iv = f(z) = x^2 - y^2 + 2xyi$$

Separating real and imaginary part

$$u = x^2 - y^2 \quad \text{and} \quad v = 2xy$$

$$\therefore \frac{\partial u}{\partial x} = 2x - 0$$

$$\Rightarrow \frac{\partial u}{\partial x} = 2x$$

$$\therefore \frac{\partial u}{\partial y} = 0 - 2y$$

$$\Rightarrow \frac{\partial u}{\partial y} = -2y$$

$$\therefore \frac{\partial v}{\partial x} = 2y \cdot 1 = 2y$$

$$\Rightarrow -\frac{\partial v}{\partial x} = -2y$$

$$\therefore \frac{\partial v}{\partial y} = 2x \cdot 1 = 2x$$

Since $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

So, the given function $f(z) = z^2$ is analytic.

Now, the derivative of $f(z)$ is

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= 2x + i(2y)$$

$$= 2(x + iy)$$

$$\Rightarrow f'(z) = 2z \text{ (Ans.)}$$

Laplace equation:

In two dimension, a second order partial differential equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is called laplace equation for the function $u = f(x, y)$.

Harmonic function:

Any function which satisfies Laplace's equation is called harmonic function.

Example:

If $u = f(x, y)$ be a function, then the harmonic will be $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Conjugate harmonic function:

If $f(z) = u + iv$ be an analytic function, where u and v satisfies Laplace's equation. Then v is called conjugate harmonic function of u , and u is called conjugate harmonic function of v .

Determine conjugate harmonic function of u :

$$dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy.$$

Determine conjugate harmonic function of v :

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy.$$

prob:

Show that the real and imaginary part of a complex function satisfies Laplace's equation.

Solution:

Let the complex function be $f(z) = u + iv$.

We know,

The Cauchy Riemann equation for $f(z) = u + iv$ is

$$\left. \begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \text{and } \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \end{aligned} \right\} \text{--- (1)}$$

partially differentiating first part of (1) w.r.t x and 2nd part of (1) w.r.t y , we get

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \text{--- (2)}$$

$$\text{and } \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x} \text{--- (3)}$$

Adding (2) and (3), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial y \partial x}$$

$$= \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial x \partial y} \left[\because \frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial y \partial x} \right]$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

So, we can say that the real part of the complex function satisfies Laplace's equation.

Again, differentiating partially first part of (1) w.r.t y and 2nd part of (1) w.r.t x , we get

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 v}{\partial y^2} \quad \text{--- (4)}$$

$$\text{and } \frac{\partial^2 u}{\partial x \partial y} = -\frac{\partial^2 v}{\partial x^2} \quad \text{--- (5)}$$

Subtracting (5) from (4), we get

$$\frac{\partial^2 u}{\partial y \partial x} - \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2}$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 u}{\partial y \partial x} - \frac{\partial^2 u}{\partial x \partial y}$$

$$= \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial^2 u}{\partial x \partial y} \left[\because \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x} \right]$$

$$\therefore \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

So, we can say that the imaginary part of the complex function satisfies Laplace's equation.

prob:

Show that the following function is harmonic.

(a) $u = y^3 - 3xy^2$ (b) $u = x^3 - 3xy^2$ (c) $u = 2xy + y^3 - 3x^2y$

(d) $u = \sin x \cosh y$ (e) $u = \sin x \sinh y$

(f) $u = \cos x \cosh y$.

Determine its harmonic conjugate and construct the corresponding analytic function $f(z)$.

Solution:

(a) Given that

$$u = y^3 - 3x^2y$$

The function u will be harmonic if

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

$$\therefore \frac{\partial u}{\partial x} = 0 - 3y \cdot 2x = -6xy$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = -6y \cdot 1 = -6y \quad \text{--- (1)}$$

$$\text{and } \frac{\partial u}{\partial y} = 3y^2 - 3x^2 \cdot 1 = 3y^2 - 3x^2$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = 3 \cdot 2y - 0 = 6y \quad \text{--- (2)}$$

Adding (1) and (2), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -6y + 6y = 0$$

$$\text{Since } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

So, the given function u is harmonicLet v be the conjugate harmonic of u .

$$\begin{aligned} \therefore dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \\ &= -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \end{aligned} \quad \left| \begin{array}{l} \text{from C-R equation} \\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \Rightarrow -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \end{array} \right.$$

$$\Rightarrow dv = (3x^2 - 3y^2)dx + (-6xy)dy \text{ which is } M dx + N dy \text{ form that means exact form.}$$

$$\int dv = \int M dx + \int (\text{terms of } N \text{ free from } x) dy + C$$

$$\Rightarrow \int dv = \int (3x^2 - 3y^2) dx + C$$

$$\therefore v = \int 3x^2 dx - \int 3y^2 dx + C$$

$$= 3 \int x^2 dx - 3y^2 \int dx + C$$

$$= 3 \cdot \frac{x^3}{3} - 3y^2 \cdot x + C$$

$$\Rightarrow v = x^3 - 3xy^2 + C$$

$\therefore$ The conjugate harmonic function $v = x^3 - 3xy^2 + C$.

Now,

The corresponding analytic function $f(z)$ is

$$f(z) = u + iv$$

$$\Rightarrow f(z) = y^3 - 3xy^2 + i(x^3 - 3xy^2 + C)$$

(d) Given that

$$u = \sin x \cosh y$$

The function u will be harmonic if

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\therefore \frac{\partial u}{\partial x} = \cos x \cosh y$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y \quad \text{--- (1)}$$

$$\therefore \frac{\partial u}{\partial y} = \sin x \sinh y$$

$$\Rightarrow \frac{\partial^2 u}{\partial y^2} = \sin x \cosh y \quad \text{--- (2)}$$

$$\left| \begin{array}{l} \frac{d}{dy} (\sinh y) = \cosh y \\ \frac{d}{dy} (\cosh y) = \sinh y \end{array} \right.$$

Adding (1) and (2), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\sin x \cosh y + \sin x \cosh y = 0$$

$$\text{since } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

So, the given function u is harmonic.

Let v be the conjugate harmonic of u .

$$\begin{aligned} \therefore dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy & \left| \begin{array}{l} \text{From C-R equation} \\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \Rightarrow -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \end{array} \right. \\ &= -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \end{aligned}$$

$\Rightarrow dv = -\sin x \sinh y dx + \cos x \cosh y dy$ which is exact form that means $M dx + N dy$ form.

$$\int dv = \int -\sin x \sinh y dx +$$

$$\therefore \int dv = \int M dx + \int (\text{terms of } N \text{ free from } x) dy + c$$

$$\Rightarrow \int dv = \int -\sin x \sinh y dx + c$$

$$\begin{aligned} \therefore v &= -\sinh y \int \sin x dx + c \\ &= -\sinh y (\cos x) + c \end{aligned}$$

$$\Rightarrow v = \cos x \sinh y + c$$

The conjugate harmonic function $v = \cos x \sinh y + c$
Now, The corresponding analytic function $f(z)$ is

$$f(z) = u + iv$$

$$\Rightarrow f(z) = \sin x \cosh y + i (\cos x \sinh y + c)$$

Adding (1) and (2), we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\sin x \cosh y + \sin x \cosh y = 0$$

Since $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

So, the given function u is harmonic.

Let v be the conjugate harmonic of u .

$$\begin{aligned} \therefore dv &= \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy & \left| \begin{array}{l} \text{From C-R equation} \\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \\ \Rightarrow -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} \end{array} \right. \\ &= -\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy \end{aligned}$$

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$$\begin{aligned} \therefore v &= -\sinh y \int \sin x dx + c \\ &= -\sinh y (\cos x) + c \end{aligned}$$

$$\Rightarrow v = \cos x \sinh y + c$$

The conjugate harmonic function $v = \cos x \sinh y + c$.
Now, The corresponding analytic function $f(z)$ is

$$f(z) = u + iv$$

$$\Rightarrow f(z) = \sin x \cosh y + i (\cos x \sinh y + c)$$